# Measure Theory: A Compact Note

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## 1 Foundations

**Definition 1.1** ($\sigma$-algebra)**.** A collection $\mathcal{F}$ of subsets of a set $\Omega$ is a **$\sigma$-algebra** if:

1. $\Omega \in \mathcal{F}$,
2. $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$ (closed under complementation),
3. $A_1, A_2, \ldots \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ (closed under countable unions).

**Definition 1.2** (Measurable set)**.** A set $A$ is **measurable** (with respect to $\mathcal{F}$) if $A \in \mathcal{F}$.

**Definition 1.3** (Borel $\sigma$-algebra)**.** The **Borel $\sigma$-algebra** $\mathcal{B}(\mathbb{R})$ is the smallest $\sigma$-algebra on $\mathbb{R}$ containing all open sets.

**Example 1.4** (Vitali set — a non-measurable set)**.** For each $r \in [0,1]$, consider the equivalence class $r + \mathbb{Q} = \{r + q : q \in \mathbb{Q}\}$. Two reals belong to the same class if and only if their difference is rational. By the axiom of choice, pick one representative from each equivalence class intersected with $[0,1]$ to form $V$. Let $\{r_k\}_{k=1}^{\infty}$ enumerate $\mathbb{Q} \cap [-1,1]$ and set $V_k = V + r_k$. The translates $\{V_k\}$ are pairwise disjoint (if $v_1 + r_j = v_2 + r_k$ then $v_1 - v_2 \in \mathbb{Q}$, so $v_1$ and $v_2$ lie in the same class, giving $v_1 = v_2$ and $r_j = r_k$) and satisfy

$$[0,1] \;\subseteq\; \bigcup_{k=1}^{\infty} V_k \;\subseteq\; [-1,2].$$

If $V$ were Lebesgue measurable, translation invariance gives $\mu(V_k) = \mu(V)$ for all $k$, and $\sigma$-additivity yields

$$1 \;\leq\; \mu\left(\bigcup_{k=1}^{\infty} V_k\right) = \sum_{k=1}^{\infty} \mu(V_k) = \sum_{k=1}^{\infty} \mu(V) \;\leq\; 3 \;(\because V_i \cap V_j = \emptyset; i \neq j).$$

This is impossible: if $\mu(V) = 0$ the sum is 0, and if $\mu(V) > 0$ the sum is $+\infty$.

**Definition 1.5** (Measurable space)**.** A pair $(\Omega, \mathcal{F})$ where $\Omega$ is a set and $\mathcal{F}$ is a $\sigma$-algebra on $\Omega$.

**Definition 1.6** (Lebesgue measure)**.** The Lebesgue measure $\mu$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is the unique measure satisfying $\mu([a,b]) = b - a$.

*Remark* 1.7. Lebesgue measure is just one example of a measure. In probability we use probability measures ($\mu(\Omega) = 1$); in counting problems, the counting measure; etc.

**Definition 1.8** (Measurable function)**.** $f\colon (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is **measurable** if $f^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{B}(\mathbb{R})$.

**Definition 1.9** ($\sigma$-additivity)**.** A set function $\mu\colon \mathcal{F} \to [0,\infty)$ is **$\sigma$-additive** if for any countable collection of pairwise disjoint sets $\{A_i\}_{i=1}^{\infty} \subset \mathcal{F}$:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

**Definition 1.10** (Measure space)**.** A triple $(\Omega, \mathcal{F}, \mu)$ where $(\Omega, \mathcal{F})$ is a measurable space and $\mu\colon \mathcal{F} \to [0,\infty)$ is a $\sigma$-additive set function with $\mu(\emptyset) = 0$.

**Definition 1.11** ($\sigma$-finiteness)**.** A measure $\mu$ on $(\Omega, \mathcal{F})$ is **$\sigma$-finite** if $\Omega = \bigcup_{i=1}^{\infty} A_i$ with $\mu(A_i) < \infty$ for all $i$. This condition is required by Fubini's theorem and Radon–Nikodym.

# 2 Modes of Convergence

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\{X_n\}, X$ random variables.

| **Mode** | **Definition** |
|---|---|
| $L^p$ convergence | $\lim_{n\to\infty} E[\lvert X_n - X\rvert^p] = 0$ |
| Almost sure (a.s.) | $P\Big(\lim_{n\to\infty} X_n = X\Big) = 1$ |
| In probability | $\forall\, \varepsilon > 0,\ \lim_{n\to\infty} P(\lvert X_n - X\rvert > \varepsilon) = 0$ |
| In distribution | $\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$ at all continuity points of $F_X$ |

$$\begin{tikzcd}
L^p \arrow[r, Rightarrow] & \textbf{in probability} \arrow[r, Rightarrow] & \textbf{in distribution} \\
& \textbf{a.s.} \arrow[u, Rightarrow] &
\end{tikzcd}$$

- $L^p \not\Rightarrow$ a.s. and a.s. $\not\Rightarrow L^p$ in general.
- In distribution $\not\Rightarrow$ in probability (unless the limit is a constant).

# 3 Core Theorems and Inequalities

## 3.1 Inequalities

**Theorem 3.1** (Markov's inequality)**.** *For a non-negative random variable $X$ and $a > 0$: $P(X \geq a) \leq E[X]/a$.*

*Usage: Crude tail bound from the mean; foundation for Chebyshev.*

**Theorem 3.2** (Chebyshev's inequality)**.** *If $E[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2$: $P(\lvert X - \mu\rvert \geq k\sigma) \leq 1/k^2$.*

*Usage: Distribution-free tail bound; used to prove the weak LLN.*

**Theorem 3.3** (Hoeffding's inequality)**.** *Let $X_1, \ldots, X_n$ be independent with $X_i \in [a_i, b_i]$ a.s. Then for $\bar{X} = \frac{1}{n}\sum X_i$:*

$$P(\lvert \bar{X} - E[\bar{X}]\rvert \geq t) \leq 2\exp\left(-\frac{2n^2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

*Usage: Exponential concentration bound; widely used in ML generalization theory (PAC learning).*

**Theorem 3.4** (Monotone Convergence Theorem (MCT))**.** *If $0 \leq f_1 \leq f_2 \leq \cdots$ are measurable and $f_n \uparrow f$ pointwise, then:*

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.$$

*Proof.* We prove the statement step by step:

1. **Upper bound.** Since $f_n \leq f$ for all $n$, monotonicity of the integral gives $\int f_n \, d\mu \leq \int f \, d\mu$ for every $n$. The sequence $\{\int f_n \, d\mu\}$ is non-decreasing ($\because f_n \leq f_{n+1}$), so its limit exists in $[0, \infty)$ and satisfies

$$\lim_{n\to\infty} \int f_n \, d\mu \leq \int f \, d\mu.$$

2. **Lower bound — setup.** It remains to show $\lim_{n\to\infty} \int f_n \, d\mu \geq \int f \, d\mu$. Recall that by definition of the Lebesgue integral for non-negative measurable functions,

$$\int f \, d\mu = \sup\left\{ \int s \, d\mu : s \text{ simple}, \ 0 \leq s \leq f \right\}.$$

So it suffices to show $\lim_{n\to\infty} \int f_n \, d\mu \geq \int s \, d\mu$ for every simple function $s$ with $0 \leq s \leq f$.

3. **Lower bound — core argument.** Fix such a simple function $s$ and fix $0 < c < 1$. Define $A_n = \{x \in \Omega : f_n(x) \geq c \cdot s(x)\}$. Since $f_n \uparrow f \geq s > c \cdot s$, every $x \in \Omega$ eventually belongs to $A_n$, so $A_n \uparrow \Omega$. Now:

$$\int f_n \, d\mu \geq \int_{A_n} f_n \, d\mu \geq c \int_{A_n} s \, d\mu.$$

The first inequality holds because $f_n \geq 0$; the second because $f_n \geq c \cdot s$ on $A_n$.

4. **Taking limits.** Write $s = \sum_{j=1}^{k} a_j \mathbf{1}_{E_j}$ where $E_j \in \mathcal{F}$. Then $\int_{A_n} s \, d\mu = \sum_j a_j \, \mu(A_n \cap E_j)$. Since $A_n \uparrow \Omega$, continuity of measure from below gives $\mu(A_n \cap E_j) \to \mu(E_j)$ for each $j$. Therefore $\int_{A_n} s \, d\mu \to \int s \, d\mu$, and so

$$\lim_{n\to\infty} \int f_n \, d\mu \geq c \int s \, d\mu.$$

Letting $c \uparrow 1$ gives $\lim_{n\to\infty} \int f_n \, d\mu \geq \int s \, d\mu$. Since this holds for every simple $s \leq f$, taking the supremum yields $\lim_{n\to\infty} \int f_n \, d\mu \geq \int f \, d\mu$.

Combining Steps 1 and 4: $\displaystyle\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.$ □

**Lemma 3.5** (Fatou's Lemma)**.** *If $\{f_n\}$ are non-negative measurable functions, then:*

$$\int \liminf_{n\to\infty} f_n \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu.$$

*Usage: "The integral of the limit $\leq$ the limit of the integrals." Used when we lack a dominating function for DCT.*

**Theorem 3.6** (Lebesgue's Dominated Convergence Theorem (DCT))**.** *If $f_n \to f$ pointwise a.e., and there exists an integrable $g$ with $|f_n| \leq g$ a.e. for all $n$, then:*

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.$$

*Proof.* We prove the statement step by step:

1. **Verifying Fatou's lemma applies.** Since $|f_n| \leq g$ a.e., the functions $g + f_n \geq 0$ and $g - f_n \geq 0$ a.e., so Fatou's lemma is applicable to both sequences. Also, $f_n \to f$ pointwise a.e. and $|f| \leq g$ a.e., so $f$ is integrable (i.e., $\int |f| \, d\mu \leq \int g \, d\mu < \infty$).

2. **Lower bound on** $\liminf$**.** Apply Fatou's lemma to the non-negative sequence $g + f_n$:

$$\int \liminf_{n\to\infty} (g + f_n) \, d\mu \leq \liminf_{n\to\infty} \int (g + f_n) \, d\mu.$$

The left side equals $\int (g + f) \, d\mu = \int g \, d\mu + \int f \, d\mu$ ($\because g + f_n \to g + f$ pointwise a.e.). The right side equals $\int g \, d\mu + \liminf_{n\to\infty} \int f_n \, d\mu$ ($\because \int g \, d\mu < \infty$ is a constant). Subtracting $\int g \, d\mu$, which is finite, from both sides:

$$\int f \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu.$$

3. **Upper bound on** $\limsup$**.** Apply Fatou's lemma to the non-negative sequence $g - f_n$:

$$\int \liminf_{n\to\infty}(g - f_n)\, d\mu \le \liminf_{n\to\infty} \int (g - f_n)\, d\mu.$$

The left side equals $\int (g-f)\, d\mu = \int g\, d\mu - \int f\, d\mu$. The right side equals $\int g\, d\mu + \liminf_{n\to\infty}(-\int f_n\, d\mu) = \int g\, d\mu - \limsup \int f_n\, d\mu$ (using $\liminf_{n\to\infty}(-a_n) = -\limsup(a_n)$). Subtracting $\int g\, d\mu$ from both sides and multiplying by $-1$:

$$\limsup_{n\to\infty} \int f_n\, d\mu \le \int f\, d\mu.$$

Combining Steps 2 and 3:

$$\int f\, d\mu \le \liminf_{n\to\infty} \int f_n\, d\mu \le \limsup_{n\to\infty} \int f_n\, d\mu \le \int f\, d\mu.$$

All inequalities are equalities, so the limit exists and equals $\int f\, d\mu$. □

# 4 Limit & Measure Theorems

**Theorem 4.1** (Fubini's Theorem)**.** *Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be $\sigma$-finite measure spaces. If $f\colon X \times Y \to \mathbb{R}$ is integrable w.r.t. the product measure, then:*

$$\int_{X\times Y} f\, d(\mu \times \nu) = \int_X \left( \int_Y f(x, y)\, d\nu \right) d\mu = \int_Y \left( \int_X f(x, y)\, d\mu \right) d\nu.$$

*Usage: Justifies swapping integration order. In ML, used for computing expectations over joint distributions.*

**Theorem 4.2** (Radon–Nikodym Theorem)**.** *Let $\mu, \nu$ be $\sigma$-finite on $(\Omega, \mathcal{F})$ with $\nu \ll \mu$ (**absolute continuity**: $\nu \ll \mu$ means that for every $A \in \mathcal{F}$, $\mu(A) = 0 \implies \nu(A) = 0$). Then there exists measurable $f \ge 0$ such that $\nu(A) = \int_A f\, d\mu$ for all $A \in \mathcal{F}$. The function $f = d\nu/d\mu$ is the **Radon–Nikodym derivative**.*

*Usage: Foundation for probability densities, likelihood ratios, and importance sampling.*

**Theorem 4.3** (Law of Large Numbers (LLN))**.** *Let $X_1, X_2, \ldots$ be i.i.d. with $E[X_1] = \mu$.*

- ***Weak:*** $\bar{X}_n \xrightarrow{P} \mu$.
- ***Strong:*** $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$ *(assuming $E[X_1^2] < \infty$).*

*Usage: Justifies sample means as estimators; underpins Monte Carlo methods and ERM.*

**Theorem 4.4** (Central Limit Theorem (CLT))**.** *Let $X_1, X_2, \ldots$ be i.i.d. with $E[X_1] = \mu$, $\mathrm{Var}(X_1) = \sigma^2 \in (0, \infty)$. Then:*

$$\sqrt{n}\,(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

*Proof.* We prove the statement step by step:

1. **Standardization.** Let $Y_i = (X_i - \mu)/\sigma$ so that $E[Y_i] = 0$ and $E[Y_i^2] = 1$. Define $S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i$. Then $S_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$, so it suffices to show $S_n \xrightarrow{d} N(0, 1)$.

2. **Characteristic function of** $Y_1$**.** The characteristic function of $Y_1$ is $\varphi_Y(t) = E[e^{itY_1}]$. By Taylor expansion (using $E[Y_1] = 0$ and $E[Y_1^2] = 1$):

$$\varphi_Y(t) = E\left[1 + itY_1 + \frac{(it)^2}{2} Y_1^2 + o(t^2)\right] = 1 + it \cdot 0 - \frac{t^2}{2} \cdot 1 + o(t^2) = 1 - \frac{t^2}{2} + o(t^2)$$

as $t \to 0$.

3. **Characteristic function of $S_n$.** Since $Y_1, \ldots, Y_n$ are independent, the characteristic function of $S_n$ factorizes:

$$\varphi_{S_n}(t) = E\left[e^{it \cdot \frac{1}{\sqrt{n}} \sum_{j=1}^n Y_j}\right] = \prod_{j=1}^n \varphi_Y\left(\frac{t}{\sqrt{n}}\right) = \left[\varphi_Y\left(\frac{t}{\sqrt{n}}\right)\right]^n.$$

4. **Taking the limit.** Substituting the expansion from Step 2 with $t$ replaced by $t/\sqrt{n}$:

$$\varphi_{S_n}(t) = \left[1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right]^n.$$

   Using the standard limit $\lim_{n\to\infty}(1 + a/n)^n = e^a$ (here $a = -t^2/2$, and the $o(1/n)$ remainder vanishes in the limit):

$$\lim_{n\to\infty} \varphi_{S_n}(t) = e^{-t^2/2}.$$

   This is the characteristic function of $N(0, 1)$.

5. **Conclusion via Lévy's continuity theorem.** Lévy's continuity theorem states: if $\varphi_{S_n}(t) \to \varphi(t)$ pointwise for all $t \in \mathbb{R}$, and $\varphi$ is continuous at $t = 0$, then $S_n$ converges in distribution to the distribution with characteristic function $\varphi$.

Since $e^{-t^2/2}$ is continuous (everywhere, in particular at 0), we conclude $S_n \xrightarrow{d} N(0, 1)$, which gives $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$. □

**Theorem 4.5** (Continuous Mapping Theorem)**.** *If $X_n \xrightarrow{d} X$ and $g$ is continuous a.e. (w.r.t. the distribution of $X$), then $g(X_n) \xrightarrow{d} g(X)$. Analogous results hold for convergence in probability and a.s.*

*Usage: Extends convergence through transformations, e.g. $X_n \xrightarrow{d} X \Rightarrow X_n^2 \xrightarrow{d} X^2$.*

**Theorem 4.6** (Cramér–Wold Theorem)**.** *$\boldsymbol{X}_n \xrightarrow{d} \boldsymbol{X}$ in $\mathbb{R}^d$ if and only if $\boldsymbol{t}^\top \boldsymbol{X}_n \xrightarrow{d} \boldsymbol{t}^\top \boldsymbol{X}$ for every $\boldsymbol{t} \in \mathbb{R}^d$.*

*Usage: Reduces multivariate convergence in distribution to univariate. Essential for proving the multivariate CLT.*

**Theorem 4.7** (Glivenko–Cantelli Theorem)**.** *Let $F_n$ be the empirical CDF of i.i.d. samples from distribution $F$. Then:*

$$\sup_{x\in\mathbb{R}} |F_n(x) - F(x)| \xrightarrow{\text{a.s.}} 0.$$

*Usage: The empirical CDF converges uniformly to the true CDF. Foundational for the Kolmogorov–Smirnov test and uniform convergence in statistical learning theory.*